



Thin-film lithium tantalate electro-optic modulator with balanced bandwidth-voltage and low DC drift

HAIWEI CHEN,^{1,2,3,†}  ZHILIN YE,^{2,4,†} KAIYAN PAN,² HEYI CAI,²
NAN JIANG,² GANG ZHAO,² SHINING ZHU,² AND XIAOPENG HU^{2,*} 

¹College of Electronic and Engineering, Suzhou Polytechnic University, Suzhou 215104, China

²National Laboratory of Solid State Microstructures, College of Engineering and Applied Sciences, and Collaborative Innovation Center of Advanced Microstructures, Nanjing University, Nanjing 210093, China

³i-Lab, Suzhou Institute of Nano-Tech and Nano-Bionics, Chinese Academy of Sciences, Suzhou, Jiangsu, China

⁴Nanzhi Institute of Advanced Optoelectronic Integration, Nanjing 211800, China

[†]These authors contributed equally to this work

*xphu@nju.edu.cn

Abstract: We demonstrate a thin-film lithium tantalate (TFLT) Mach-Zehnder modulator with balanced bandwidth-voltage and low DC drift. Precise waveguide geometry optimization yields a 3-dB bandwidth (BW) > 67 GHz and a half-wave voltage-length product ($V_{\pi}L$) of 2.4 V·cm, resulting in a higher BW/V_{π}^2 compared to the previously reported TFLT modulators. The device also shows excellent operational stability, with its V_{π} remaining stable across a wide range of driving frequencies and power drift below 1 dB over 60 minutes. Furthermore, we develop what we believe is a novel and stable measurement technique which employs a cascaded dual-ring modulation structure to quantitatively assess the DC drift, confirming the long-term viability of TFLT modulators for high-speed, stable, and low-power integrated photonics.

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1. Introduction

Thin-film lithium niobate (TFLN) electro-optic (EO) integrated circuits have been rapidly advancing and showing great promise for furthering optical science and technology in recent years [1–5]. Similarly, thin-film lithium tantalate (TFLT), which belongs to the same family as LN, has also garnered significant attention due to its outstanding EO performance and robust stability [6–10]. Recently, driven by its large Pockels coefficient (~30 pm/V) [8,11] and broad transparent window (280–5500 nm) [12], TFLT has achieved a breakthrough, enabling EO modulators operating near the visible spectrum [13]. Furthermore, due to its high optical damage threshold [14,15] (240 MW/cm²), TFLT modulators are highly promising for next-generation high-power photonic systems, enabling optical power handling up to 630.9 mW [16]. It is well known that the half-wave voltage (V_{π}) and bandwidth (BW) are the critical metrics for EO modulators, directly determining the achievable data rate and power consumption in high-baud-rate optical transmission systems [6,17]. Similar to LN, TFLT EO modulator with a 3-dB BW exceeding 110 GHz has been demonstrated [7]. Moreover, the half-wave voltage-length product ($V_{\pi}L$) has been reduced to as low as 0.65 V·cm [13], enabling compatibility with CMOS driving voltages. However, the corresponding 3-dB BW for the EO modulator with a low V_{π} is limited to merely 5 GHz [13], because focusing solely on minimizing V_{π} neglects microwave loss and velocity mismatch that limit modulation bandwidth. Extending the modulation length reduces V_{π} , but concurrently introduces increased microwave loss and velocity mismatch, thereby degrading the 3-dB BW. Consequently, achieving a balance between BW and V_{π} is extremely important in photonic integrated circuit design [6,17,18], especially in high-speed and low-power systems. Since the required driving power of EO modulator is proportional to the V_{π}^2 [19], we can define

a figure of merit (FoM) equal to BW/V_π^2 to simultaneously reveal the connection between the modulation bandwidth and power consumption of the device [10,18]. The larger the value of BW/V_π^2 , the better the balance between the bandwidth and voltage. Therefore, maximizing the BW/V_π^2 is one of the important considerations for achieving high data rates and low power consumption.

In addition to exhibiting EO properties comparable to or even surpassing those of LN, TFLT photonics is strongly driven by its lower birefringence and weakened charge-transport effects [9,20]. These effects are known to induce strong EO relaxation [21,22], which degrades the direct current (DC) stability of EO circuits, presenting one of the major challenges for TFLN photonics. In practice, the performance of TFLN EO modulators is often sensitive to fabrication variations and substrate quality, leading to static operating point drift [1,23–25]. Fortunately, TFLT EO modulators intrinsically exhibit ultra-low DC drift, effectively compensating for this limitation. Several studies have proved that TFLT Mach-Zehnder modulator (MZM) has superior stability performance compared to TFLN [6,7,9,16]. However, these attempts are only achieved by indirectly assessing the DC drift through monitoring optical power fluctuations. These methods are subject to various instabilities (light source, detector, and system drift), leading to unreliable results and compromised accuracy, which severely hinders further device optimization and application. Therefore, it is imperative to develop a more stable and precise measurement technique to provide reliable experimental data for in-depth research and to resolve the issue of DC drift in integrated devices.

Here, we design and fabricate a TFLT MZM operating at 1550 nm, yielding a 3-dB BW exceeding 67 GHz and a low $V_\pi L$ of 2.4 V·cm. Based on the precise optimization of the waveguide structure within a limited length, a balance between bandwidth and voltage has been achieved in TFLT modulator. To the best of our knowledge, among the TFLT modulators reported so far, the device has the optimal FoM of BW/V_π^2 . Crucially, to investigate long-term performance, we develop an apparently novel and stable measurement technique, namely using a cascaded dual-ring modulation structure to continuously and precisely monitor the changes in the EO response caused by DC drift. Experimental results confirm that the stability of TFLT modulator under same conditions significantly surpasses that of counterpart TFLN.

2. Design and fabrication

Figure 1(a) illustrates the 3D structural schematic of the designed Mach-Zehnder (MZ) EO modulator. The device is fabricated on an *x*-cut lithium tantalate on insulator wafer, employing the coplanar waveguide electrodes. The wafer stack consists of a 600 nm thin-film LT, a 4.7 μm thick silicon dioxide, and a silicon substrate. The modulator incorporates a ridge waveguide with a height of 300 nm and a sidewall angle of 69°. The splitter and combiner are designed as multimode interference (MMI) structures. Considering practical fabrication constraints, the total length of the modulation section is set to 7.5 mm, with an optimized electrode gap between the signal and ground electrode maintained at 6 μm . It is well-established that the V_π is critically dependent on the modal overlap factor (Γ_m) between the RF electric field and the optical field, while the operational BW is ultimately limited by velocity matching and microwave loss. For a fixed electrode gap, increasing the waveguide width generally decreases the Γ_m , thereby increasing the V_π [19]. For BW/V_π^2 , there will be an optimal width due to the balance between velocity matching and microwave loss [26]. Initially, wider waveguides improve velocity matching, significantly enhancing the BW. The BW/V_π^2 gradually increases due to the fact that the increase rate of BW outpaces that of V_π^2 . However, beyond the optimal width, further widening leads to increased microwave loss, causing the BW to saturate while V_π continues to rise. Consequently, after the BW/V_π^2 reaches a maximum at an optimal width, there is a slight downward trend. Figure 1(b) shows the simulated trend of V_π and BW/V_π^2 as the waveguide width changes from 1.6 μm to 2.6 μm , confirming that the optimal value is achieved when the top

width of ridge waveguide is $2.2\ \mu\text{m}$. It is worth noting that the ridge waveguide with a $2.2\ \mu\text{m}$ top width is only used within two modulating arms to enhance modulation efficiency. Crucially, before entering the MMI, the waveguide width is adiabatically tapered down to $1\ \mu\text{m}$, which is designed to be strictly single-mode. Therefore, optical signal transmission in such a waveguide is theoretically impossible to excite higher-order modes, and it will not affect the extinction ratio of EO modulation. The simulated RF effective index (red line) and characteristic impedance (blue line) are shown in Fig. 1(c). The characteristic impedance is designed to be close to $50\ \Omega$.

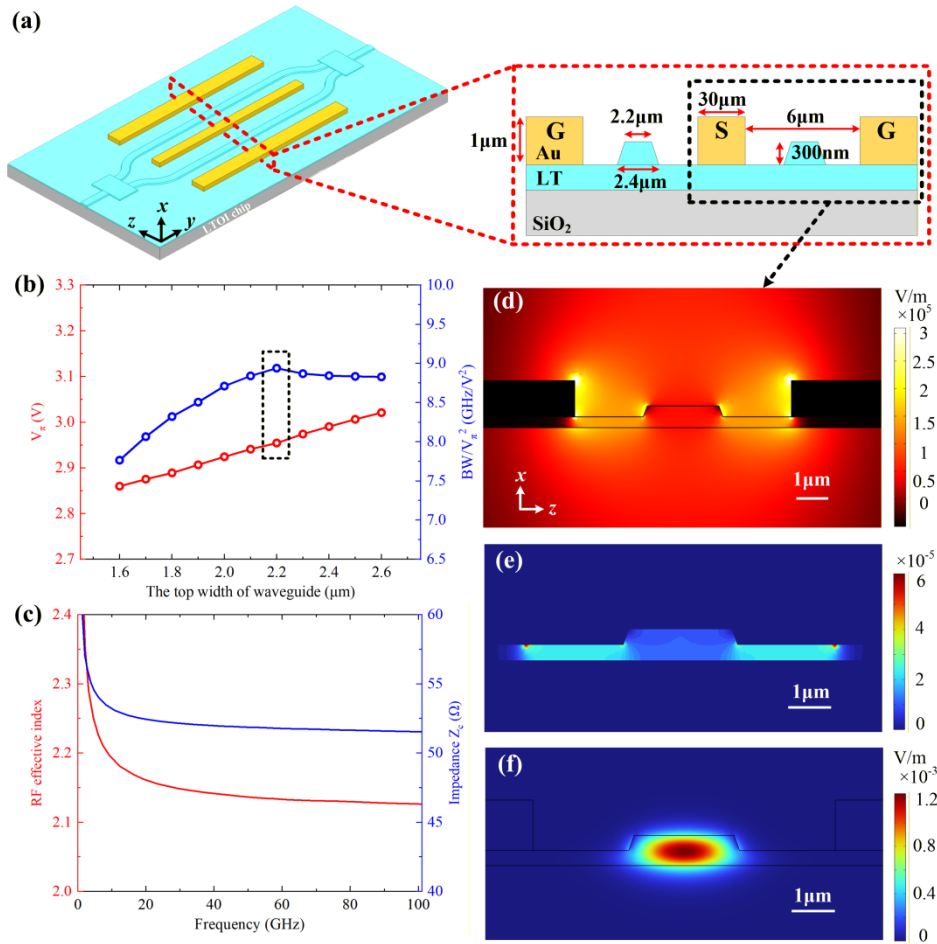


Fig. 1. (a) Schematic diagram of the TFLT Mach-Zehnder electro-optic modulator. Inset: Cross-sectional schematic diagram of the modulation part. (b) Changes in V_{π} and BW/V_{π}^2 with the variation of the top width of ridge waveguide. (c) Simulation results of the RF effective refractive index (red line) and characteristic impedance (blue line). (d) Simulation electric field cross-section distribution (E_z) when the voltage between the two electrodes being 1 V. (e) Distribution of the refractive index variation within the TFLT waveguide. (f) Relative electric field amplitude cross-sectional distribution for the low-order TE mode.

The optical and electrical modes of the modulator is simulated by using a multiphysics field solver based on the finite element method at a wavelength of 1550 nm. As depicted in Fig. 1(d), the simulated electric field distribution in the cross-section, assuming an applied voltage of 1 V across the electrodes, reveals excellent field overlap with the TFLT and strong field intensity within the waveguide region. Based on the electric field distribution, the resulting refractive

index change (Δn) distribution within the TFLT waveguide can be further calculated using the relation $\Delta n = E_z n_e^3 r_{33}/2$, as depicted in Fig. 1(e). Furthermore, the transverse distribution of the fundamental TE mode electric field amplitude is simulated in Fig. 1(f), which clearly illustrates the high optical confinement within the waveguide for the TE mode without mode crossings.

Figure 2(a) shows the fabricated TFLT MZM, which incorporates two MMIs located at either end. The fabrication process is based on chip-level processing, primarily involving electron beam lithography (EBL), ion beam etching (IBE), and electron beam evaporation (EBE). Initially, EBL is used to define the mask for etching the waveguide. Subsequently, IBE is applied to etch the MZM waveguide structure. A crucial step involves removing residual deposition, which is attached on the waveguide sidewalls due to the physical etching of TFLT by argon ions. The residue is cleared using an alkaline solution consisting of KOH and H₂O₂ (3:1), as illustrated in the inset of Fig. 2(a). The CPW gold electrodes are then fabricated via EBE deposition followed by a lift-off process. Finally, the chip edges are cleaved to facilitate subsequent fiber-to-chip coupling with lensed fibers.

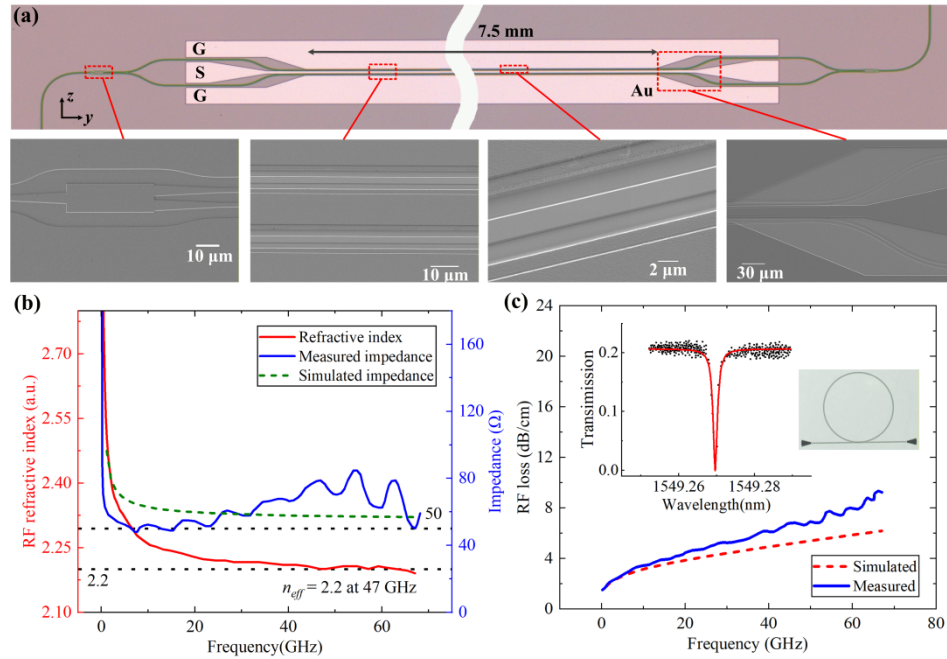


Fig. 2. (a) Optical microscope images of the 7.5 mm TFLT EO modulator. The crystal axis and electrode details are labeled. The insets: The scanning electron microscope (SEM) images show the key components of the TFLT modulator, including the MMI beam splitter, TFLT waveguide, the CPW electrode, and the ground-signal-ground (GSG) structure. (b) The blue solid line and the green dashed line represent the measured and simulated characteristic impedances, respectively. The measured RF effective index (red line) is $n_{eff} = 2.2$ at 47 GHz. (c) The simulated and measured results of the RF loss. Inset: The optical transmission spectrum of the TFLT microring resonator, with the measured results (black dots) and the Lorentz fitting curve (red line). The propagation loss of the single-mode waveguide is less than 0.5 dB/cm.

The performance of EO modulators based on TFLT is critically dependent on the characteristic impedance, the RF signal loss, and the velocity matching between the optical signal and the RF signal propagation speeds. Figure 2(b) presents the measured characteristic impedance and RF effective refractive index of the fabricated MZM. By optimizing the waveguide geometry

and choosing appropriate electrode gap, we effectively reduced the RF effective index, which enabled close alignment with the group index of the optical mode ($n_g = 2.2$). Excellent matching between the RF effective index and the group index is observed at 47 GHz (red line). Furthermore, Fig. 2(b) confirms that the electrode impedance remains close to the design target of 50 Ω (blue line). Figure 2(c) compares the measured RF loss curves against the simulated results across various frequencies. The inconsistencies between measured and simulated RF loss curves arise from fabrication tolerances (such as a slight deviation between the actual electrode size and the designed size). Despite these discrepancies, the measured trend agrees well with the simulation. The RF loss curve follows the relationship $\alpha = \alpha_0 f^{-1/2}$, where α_0 represents the frequency-independent characteristic RF loss coefficient, calculated to be 0.74 dB·cm⁻¹·GHz^{-1/2}. The inset of Fig. 2(c) shows the measurement of the waveguide propagation loss using a microring resonator. Through calculation, the single-mode propagation loss for the waveguide is determined to be approximately 0.5 dB/cm.

3. Result and discussion

We first perform a static measurement of the V_π for the fabricated TFLT MZM. Considering that the MZM waveguides are fabricated on an *x*-cut TFLT and light propagates along the *y*-direction, we consistently employ TE polarization during the experiment to fully utilize the largest electro-optic coefficient (r_{33}) of LT. The input light at 1550 nm is coupled into and out of the modulator using a setup equipped with two lens-type optical fibers. For a long MZM, a triangular wave with a peak-to-peak voltage of 8 V and a frequency of 5 kHz is applied across the electrodes. The measured V_π of the TFLT modulator at 1550 nm is 3.2 V, resulting in a corresponding $V_\pi L$ of 2.4 V·cm, as shown in Fig. 3(a).

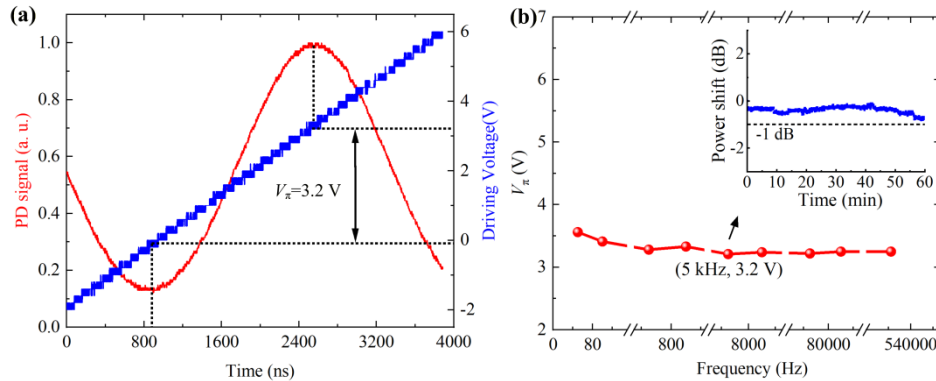


Fig. 3. (a) The measured V_π of the TFLT modulator is 3.2 V. (b) The variation of the V_π corresponding to different driving frequencies from 50 Hz to 500 kHz without changing the input optical power. Inset: The relative output power intensity with the voltage set to the second quadrature point.

Next, we investigate the influence of varying driving frequencies on the V_π of the TFLT MZM. Unlike previously reported TFLN modulators [20,21], which often exhibit strong frequency dependence due to the presence of DC drift, the TFLT MZM maintains a stable V_π of 3.2 V across a wide frequency range from 50 Hz to 500 kHz without altering the input optical power, as shown in Fig. 3(b). To further evaluate the operational stability of TFLT MZM, we measured the output power drift when the voltage is set to the second quadrature point. To exclude coupling instability, we measure the MZM transmission curve both before and after the stability test. The two curves are nearly identical, confirming stable optical coupling. As illustrated in the inset of Fig. 3(b), the drift remains within 1 dB over a 60-minute period. The excellent stability

of TFLT MZM is critical for long-distance high-speed optical communication systems, as it mitigates power fluctuations and bit error rates in the link [27]. Specifically, the low power drift minimizes the need for frequent power equalization, thereby reducing system complexity and power consumption.

Subsequently, we characterize the small-signal EO BW of the fabricated MZM. The EO response S_{21} is measured using a 67 GHz photodetector. As shown in Fig. 4(a), the 3-dB BW corresponding to the measured response curve (blue line) exceeds 67 GHz. Although the TFLT MZM maintained an EO response up to 67 GHz, there is still room for improvement in the performance of EO response. The primary limitation stems from the overly long probing region, which accounts for approximately 10% of the total modulator length. This results in two direct consequences: First, while the EO response exhibits a relatively flat trend at lower frequencies (<45 GHz), it shows pronounced fluctuations in the range from 45 to 67 GHz as shown in Fig. 4(a). Second, the characteristic impedance fluctuates significantly in the high-frequency band (>45 GHz), failing to maintain the ideal 50 Ω (as seen in Fig. 2(b)). Figure 4(b) shows the measured and simulated electrical reflection S_{11} of the gold CPW. At lower frequencies, the S_{11} is below -15 dB, revealing that good impedance matching is achieved. However, the opposite trend is observed at high frequencies. The overall trend is consistent with the variation of the characteristic impedance, further confirming that the probe region has a significant impact on the modulation bandwidth. In the next design iteration, these high-frequency ripples could be eliminated by shortening the probing region and optimizing the transition structure, which will significantly enhance the bandwidth flatness and further extend the modulation bandwidth. Furthermore, the optimization will have a negligible effect on V_{π} , since V_{π} is mainly determined by the modulation area where the RF field interacts with the optical mode.

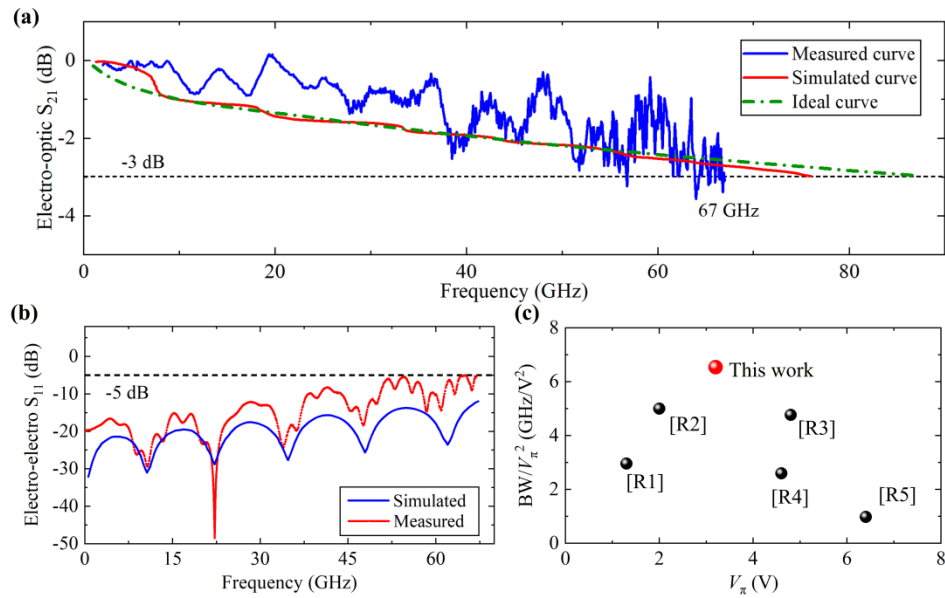


Fig. 4. (a) The measured 3-dB EO bandwidth (blue line), comprehensive simulation curve (red line), and ideal theoretical curve with the optimized probing area (green line). (b) The measured and simulated electrical reflection S_{11} of the gold electrode. (c) Comparison of the bandwidth-voltage performance of TFLT reported in different references. R1 to R5 represent the Refs. [13,28,7,16] and [8], respectively. The operation wavelength of the modulator R1 is 737 nm.

As shown in Fig. 4(c), we compare the bandwidth-voltage performance of most previously reported devices. Operating at a relatively low voltage, our device achieves the maximum FoM of BW/V_π^2 (6.8 GHz/V^2) among the currently reported TFLT modulators. The moderate discrepancy between the simulated and experimental value of FoM in Fig. 1(b) and Fig. 4(c) is mainly due to fabrication tolerances, including deviations in electrode dimensions and waveguide geometry variations. As shown in Table 1, a comparison of our modulator designs with the state-of-the-art and measured TFLN and TFLT modulators with regular CPW electrodes is presented. The result strongly validates that the precise waveguide structure design exhibits high balance performance in simultaneously achieving high energy efficiency and broad bandwidth capability. In the future, we will also explore the synergistic optimization scheme that comprehensively considers the effects of waveguide width and modulation length on V_π and BW to further enhance the FoM of BW/V_π^2 .

Table 1. Performance of the state-of-the-art TFLN and TFLT modulators with regular CPW electrodes

Reference	V_π (V)	$V_\pi \cdot L$ (V·cm)	Bandwidth (GHz)	BW/V_π^2 (GHz/V ²)	Power shift (dB)
TFLN [4]	0.9	2.7	33	4.5	/
TFLN [29]	6.6	3.3	170	3.9	/
TFLT [7]	4.8	2.9	110	4.7	3 dB
TFLT [13]	1.3	0.65	5	2.9	2 dB
Our work	3.2	2.4	67	6.8	< 1 dB

To further verify the long-term stability performance of TFLT EO modulation, we propose a measurement method for quantifying the DC drift in TFLT EO devices. As shown in Fig. 5(a), the method utilizes a cascaded dual-ring resonator structure. The resonant wavelength produced by the microring without modulation electrodes (close to the straight waveguide) does not change with whether the entire structure is powered on or off, which is used as the initial reference point. Both the gap between the straight waveguide and the microring, as well as the gap between the two microrings, are set to $0.5 \mu\text{m}$. In the modulation region of one microring resonator, the gap between the metal electrode and the waveguide is $2.4 \mu\text{m}$. The insets of Fig. 5(a) display the SEM images of the key components of the fabricated cascaded EO modulation structure. First, we apply an external electric field to the modulation section and use a data acquisition system to record the output optical signal from the resonators in real-time. Subsequently, based on the collected optical signals, we extract the wavelength shifts ($\Delta\lambda$) of the microring resonant wavelength over different durations. As shown in Fig. 5(b), the initial transmission spectra of the TFLT microring modulator under unloaded voltage (red line) and applied voltage (blue line) are acquired. With no voltage applied, two points are selected as basepoint 1 and basepoint 2 from a continuously scanned transmission spectrum. Upon applying the voltage, the initial shift of the microring resonant wavelength is recorded as the initial value reflecting the refractive index change due to the EO effect, denoted as $\Delta\lambda_1$ and $\Delta\lambda_2$. Thereafter, the transmission spectrum is recorded every few minutes. By continuously monitoring the variation in resonance wavelength, the changing trend of the EO response value due to the presence of DC drift can be clearly reflected. It is worth noting that the presence of multiple resonance wavelength points without applied voltage or with applied voltage is due to the difference in coupling strength between the straight waveguide and the microring, and between the two microrings themselves [3].

As shown in Fig. 5(c), the curves track the change in the EO response corresponding to the two initial basepoints over 2 hours without altering any other conditions for the TFLT microring modulator. Evidently, the EO response values of the cascaded TFLT microring only fluctuate slightly around the mean value over time, approximately $\pm 4\%$. Calculation reveals that

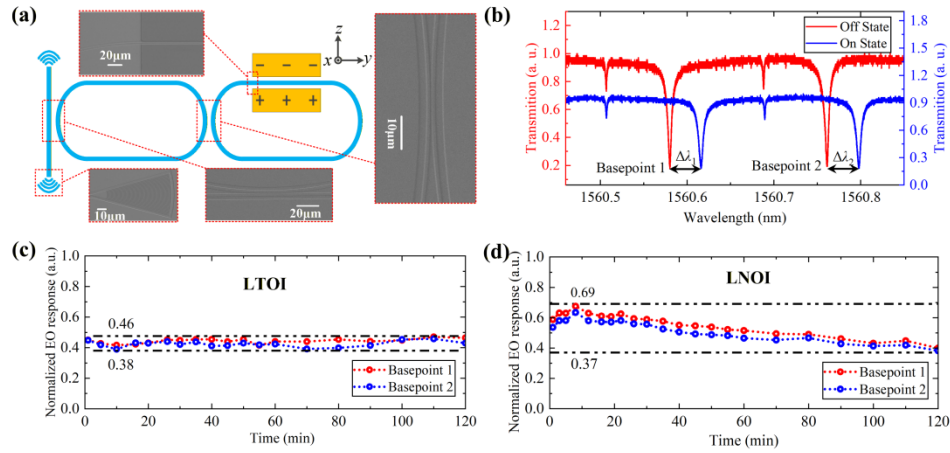


Fig. 5. (a) Schematic diagram of the cascaded dual-ring structure for monitoring DC drift. The insets: The SEM images show the key components of the fabricated cascaded EO modulation structure. The input voltage applied to the microring modulator is 8 V. (b) Transmission spectra of the TFLT microring modulator under unloaded voltage (red line) and applied voltage (blue line). Based on the collected optical signals, the wavelength variations of the microring resonant under different durations are extracted. The first wavelength shift is defined as basepoint 1, while the second is defined as basepoint 2. (c) The EO response curves corresponding to the two basepoints of the TFLT microring modulator within 2 hours without changing other conditions. (d) The EO response curves correspond to the counterpart TFLN microring modulator that is fabricated using exactly the same process and tested with the same method. Both EO response variations are normalized using the same reference point of 1.0 GHz/V.

the standard deviation and the relative variation coefficient are 0.026 and 5.8%, respectively, demonstrating the excellent long-term stability performance for the TFLT microring modulator. Similarly, the EO response curve for the counterpart TFLN microring modulator, fabricated using the exact same process and tested with the same method, is shown in Fig. 5(d). Unlike the TFLT, the response of the TFLN microring modulator exhibits a clear decreasing trend after reaching its peak over time. Within 2 hours, the variation range reaches 32%, with a tendency to continue decreasing thereafter. The stark contrast in EO response curves can be attributed to the intrinsic material properties. The superior EO stability of the TFLT modulator arises from the higher migration activation energy of Li ions in LT compared to LN [9,21]. The higher activation energy binds Li ions more strongly to their lattice sites, requiring greater electric field energy to displace them. In summary, the method innovatively converts the measurement of DC drift into the monitoring of microring resonant wavelength via the cascaded dual-ring structure, significantly improving test long-term stability and precision while simplifying system complexity. Compared to traditional power-dependent measurements [6,7,9,16], the technique more accurately reflects the true influence of the electric field on the refractive index, making it less susceptible to optical power fluctuations.

4. Conclusion

In conclusion, we successfully design and fabricate a high-performance, stable TFLT MZM by optimizing the waveguide geometry. The device achieves a 3-dB BW exceeding 67 GHz while maintaining a low $V_{\pi}L$ of 2.4 V·cm, leading to a high FoM of BW/V_{π}^2 compared to the reported TFLT modulators. The theoretical and experimental BW/V_{π}^2 are in good agreement, fully demonstrating that a balance has been achieved between bandwidth and voltage. Further,

the excellent operational stability of the TFLT MZM is confirmed through a stable V_π across wide driving frequencies from 50 Hz to 500 kHz and an output power drift of less than 1 dB over 60 minutes. Equally importantly, we also develop what we feel is a novel and stable measurement technique which employs a cascaded dual-ring modulation structure for quantifying the DC drift in EO devices, confirming that the TFLT modulator exhibits superior long-term stability performance compared to the counterpart TFLN device. Overall, these results provide significant reference value for the optimized design of EO modulators and quantitatively analyzing DC drift in integrated photonics.

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Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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